Congruences and last digits

High school · number theory · Worked solutions · 03

Exercise 1. Find the remainder of -37 modulo 8.

Solution. $-37 = 8 \times (-5) + 3$, so the remainder is 3.

Exercise 2. Find the last digit of 7^{2026}.

Solution. The cycle is 7, 9, 3, 1 and $2026 \equiv 2 \pmod 4$. The last digit is 9.

Exercise 3. Prove that an integer square is never congruent to 2 modulo 3.

Solution. An integer is congruent to 0, 1 or 2. The corresponding squares are 0, 1 and $4 \equiv 1$ modulo 3.

Exercise 4. Solve $3x \equiv 1 \pmod 7$.

Solution. $3 \times 5 = 15 \equiv 1$. Multiplying by 5 gives $x \equiv 5 \pmod 7$. Conversely, every such value works.

Problem · Investigate, justify, conclude. An integer N is congruent to 2 modulo 3 and to 3 modulo 5. Write $N = 3k + 2$, determine k modulo 5, then give all solutions modulo 15. Which solutions from 0 to 100 inclusive are even?

Solution. $3k + 2 \equiv 3 \pmod 5$, so $3k \equiv 1$ and $k \equiv 2 \pmod 5$, since $3 \times 2 \equiv 1$. Thus $N = 15t + 8$. In $[0, 100]$ the values are $8, 23, 38, 53, 68, 83, 98$. The even values are $8, 38, 68, 98$. Substitution verifies both congruences for every integer t.